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LOW PRESSURE RELIEF VALVE

Abstract

A disc for a pressure relief valve includes a disc body having a core defining a central axis of the disc, and a flange flexibly coupled to the core, the flange defining a seat surface of the disc. The seat surface is axially movable relative to the core.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. provisional application Ser. No. 63/560,835 filed Mar. 4, 2024, the contents of which are incorporated by reference.

BACKGROUND

[0002] Pressure relief valves are used in many industrial applications to prevent fluid systems from reaching undesirable high pressures. Such valves commonly include a nozzle with a valve seat that is normally closed by a slidable disc. The disc is typically biased against the valve seat in the closed configuration by a compression spring, fluid pressure, or both. When a pressure in the nozzle passage exceeds a predetermined set pressure, the disc moves against the biasing force away from the valve seat, thereby opening the valve. Conversely, when pressure in the nozzle passage decreases by a certain amount to a reseal pressure, the biasing force moves the disc back to the closed configuration.

[0003] For relief valves with low set pressures (e.g., 50 PSI or less), the biasing force on the disc must be relatively low so the disc can overcome the biasing force and open at the low threshold. However, the disc may not properly engage (and seal) the valve seat in the closed configuration if the biasing force that presses the disc against the valve seat is low. Moreover, increasing the biasing force so the disc properly engages the valve seat can prevent the disc from opening at its low set pressure.

[0004] An object of the present disclosure is to provide a disc for a relief valve that can facilitate proper engagement with the valve seat, particularly for low pressure relief valves with low set pressures and low biasing forces acting on the disc.

BRIEF SUMMARY OF THE INVENTION

[0005] The following presents a simplified summary of example embodiments of the invention. This summary is not intended to identify critical elements or to delineate the scope of the invention.

[0006] In accordance with a first aspect, a disc for a pressure relief valve includes a disc body having a core defining a central axis of the disc, and a flange flexibly coupled to the core, the flange defining a seat surface of the disc. The seat surface is axially movable relative to the core.

[0007] In one example, the flange is resiliently biased toward an unloaded configuration relative to the core, wherein the seat surface extends substantially radial to the central axis in the unloaded configuration.

[0008] In another example, the disc body includes polytetrafluoroethylene.

[0009] In yet another example, an axial seat stiffness of the flange is about 500 lb/in to about 1500 lb/in.

[0010] In still yet another example, the flange includes a distal portion that defines the seat surface, and a proximal portion that flexibly couples the distal portion to the core, wherein a ratio of a smallest axial thickness of the proximal portion to a largest axial thickness of the distal portion is about 0.20 to about 0.32.

[0011] In another example, the flange extends radially outward from the core.

[0012] In yet another example, an outer diameter of the flange is less than an outer diameter of the core, wherein a difference between the outer diameter of the flange and the outer diameter of the core is at least 0.003 inches.

[0013] In still yet another example, the flange includes a beveled surface that extends radially outward from the seat surface, wherein an angle between a direction of the beveled surface and a direction of the seat surface is about 10° to about 20°.

[0014] In another example, the flange is resiliently biased toward an unloaded configuration relative to the core, wherein an angle between the beveled surface and a radial direction of the disc in the unloaded configuration is about 10° to about 20°.

[0015] In yet another example, the disc body defines an annular groove above the flange, wherein the annular groove is defined at least partially by the flange.

[0016] In still yet another example, the disc body defines a recess, wherein the seat surface extends outward from an outer perimeter of the recess.

[0017] In another example, an edge between the outer perimeter of the recess and an inner

perimeter of the seat surface has a radius of curvature of about 0.020 inches to about 0.040 inches.

[0018] In yet another example, a disc assembly includes the disc and a disc holder defining a disc cavity that accommodates the core of the disc.

[0019] In still yet another example, the disc holder includes a frustoconical skirt surface extending outward from an outer perimeter of the disc cavity.

[0020] In another example, the disc cavity further accommodates the flange of the disc.

[0021] In yet another example, an outer diameter of the flange is less than an outer diameter of the core, wherein a difference between the outer diameter of the flange and the outer diameter of the core is at least 0.003 inches.

[0022] In still yet another example, the disc holder includes a holder surface that extends outward from an outer perimeter of the disc cavity, wherein an annular gap is defined between the flange of the disc and the holder surface of the disc holder.

[0023] In another example, the holder surface extending substantially radial to the central axis.

[0024] In yet another example, the disc holder includes a frustoconical skirt surface extending outward from an outer perimeter of the holder surface.

[0025] In still yet another example, a pressure relief valve includes the disc assembly and a nozzle defining a bore, an opening at an end of the bore, and a seat surface that surrounds a perimeter to the opening. The disc assembly is translatable between an open configuration in which the seat surface of the disc is spaced from the seat surface of the nozzle, and a closed configuration in which the seat surface of the disc engages the seat surface of the nozzle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The above and other features, aspects, and advantages of the present application are better understood when the following detailed description of the present application is read with reference to the accompanying drawings, in which:

[0027] FIG. 1 is a cross-sectional view of an example first embodiment of a relief valve;

[0028] FIG. 2 is an enlarged view of the first embodiment;

[0029] FIG. 3 is a cross-sectional view of a disc of the first embodiment, showing various dimensions of the disc;

[0030] FIG. 4 is a cross-sectional view of an example second embodiment of a relief valve; and

[0031] FIG. 5 is an enlarged view of the second embodiment.

DETAILED DESCRIPTION

[0032] The following is a detailed description of illustrative embodiments of the present application. As these embodiments of the present application are described with reference to the aforementioned drawings, various modifications or adaptations of the methods and or specific structures described may become apparent to those skilled in the art. All such modifications, adaptations, or variations that rely upon the teachings of the present application, and through which these teachings have advanced the art, are considered to be within the spirit and scope of the present application. Hence, these descriptions and drawings are not to be considered in a limiting sense as it is understood that the present application is in no way limited to the embodiments illustrated.

[0033] Moreover, certain terminology is used herein for convenience only and is not to be taken as a limitation. For example, relative directional terms herein such as “upper”, “lower”, and the like are used for convenience when describing the embodiments as oriented in the drawings. However, such orientation(s) are arbitrary, and different orientations can result rearranging which feature might be considered the “upper” or “lower,” etc.

[0034] Turning to FIG. 1, a first embodiment of an example pressure relief valve **10** is shown for

relieving pressure in a fluid system. The relief valve **10** includes a nozzle **12**, which is a generally tubular body defining a bore **14** with an opening **18** at its upper end. The relief valve **10** further includes a disc assembly **22** that is movable relative to the nozzle **12** to selectively open and close the opening **18** of the nozzle **12**.

[0035] The disc assembly **22** includes a disc holder **26** that defines a cylindrical disc cavity **28** and a frustoconical skirt surface **30** extending radially outward and downward from a lower perimeter of the disc cavity **28**. The disc assembly **22** also includes an elongated shaft **34** that extends into the disc cavity **28**, and a disc **36** that is arranged within the disc cavity **28** and coupled to the lower end of the shaft **34**. The disc **36** thereby secures the disc holder **26** to the shaft **34**. In the present example, the disc **36** is fixed to the shaft **34**. In other examples, the disc **36** can be movably (e.g., pivotally) coupled to the lower end of the shaft **34**, allowing the disc **36** and disc holder **26** to be slightly movable relative to the shaft **34** to account for potential misalignments between the disc assembly **22** and nozzle **12** during operation.

[0036] FIG. **2** shows an enlarged view of the relief valve **10**, while FIG. **3** shows a detailed view of various dimensions for the valve's disc **36**. The nozzle **12**, disc holder **26**, shaft **34**, and disc **36** are all arranged to be coaxial with a central axis X of the relief valve **10** (see FIG. **1**). In other words, the central axis X shown in FIG. **1** corresponds to the central axes of the nozzle **12**, shaft **34**, disc holder **26**, and disc **36**. Moreover, the nozzle **12**, shaft **34**, disc holder **26**, and disc **36** all extend symmetrically about the central axis X, such that their cross-sections shown in FIG. **1** extend circumferentially and continuously about the central axis X. Notably, the central axis X defines axial and radial directions of the relief valve **10** and its features described herein. The central axis X also defines inward and outward directions of the relief valve **10** and its features, such that outward features are further away from the central axis X in the radial direction as compared to inward features.

[0037] The nozzle **12** has an annular seat surface **44** that extends radially outward from the perimeter of its opening **18**. The disc assembly **22** is translatable along the central axis X between an open configuration (shown in FIG. **1**) in which the disc **36** is axially spaced from the seat surface **44**, and a closed configuration in which the disc **36** engages the seat surface **44**. Moreover, the disc assembly **22** is biased toward the closed configuration via one or more biasing elements. For example, the relief valve **10** can include a compressed spring (not shown) that applies axial force against the disc assembly **22** in a direction toward the seat surface **44**. In addition or alternatively, the disc assembly **22** may be biased toward the seat surface **44** via gravity, magnetic force, and/or some other feature that applies biasing force to disc assembly **22**.

[0038] When the disc assembly **22** is in the closed configuration and fluid pressure within the nozzle **12** exceeds a predetermined set pressure, the pressurized fluid will act against disc **36** of the disc assembly **22** and overcome its biasing force, causing the disc assembly **22** to translate axially (against the biasing force) away from the seat surface **44** to the open configuration shown in FIG. **1**. This will permit the pressurized fluid within the nozzle **12** to escape via the opening **18**. The pressurized fluid will continue to act against the disc **36** and frustoconical skirt surface **30** of the disc holder **26** as it leaves nozzle **12**, thereby maintaining the disc assembly **22** in the open configuration. However, if the fluid pressure drops below a predetermined reseal pressure, the biasing force against the disc assembly **22** will overcome the fluid force of the lower-pressure fluid, causing the disc assembly **22** to translate axially toward the seat surface **44** to the closed configuration.

[0039] Preferably, the disc **36** of the disc assembly **22** will engage the seat surface **44** of the nozzle **12** in the closed configuration to form a continuous seal around the opening **18** of the nozzle **12**, thereby inhibiting fluid within the nozzle **12** from escaping between the disc **36** and seat surface **44**. In conventional relief valves, this can typically be achieved by applying sufficient biasing force against the disc assembly, such that the disc presses and conforms against the seat surface to yield a continuous seal. However, as discussed above, the biasing force for relief valves with low set

pressures (e.g., 50 PSI or less) must be relatively low so the disc can overcome the biasing force and open at the low threshold. Thus, the disc **36** of the present embodiment is configured to facilitate proper engagement with the seat surface **44**, particularly in low-pressure applications with relatively low biasing forces acting on the disc assembly **22**.

[0040] More specifically, the disc **36** comprises a single, monolithic body **50** of material. Preferably, the disc body **50** comprises Teflon (i.e., polytetrafluoroethylene), although other materials are possible including, for example, silicone, neoprene, polychlorotrifluoroethylene, etc. The disc body **50** comprises a core **52** that is coaxial with the central axis X. The disc body **50** also includes an annular flange **54** that is flexibly coupled to the core **52** and defines an annular seat surface **58** for engagement with the seat surface **44** of the nozzle **12** in the closed configuration. The core **52** and flange **54** of the disc body **50** define an annular groove **60** just above the flange **54** that is coaxial with the central axis X and extends inward from an outer perimeter of the disc body **50**.

[0041] The flange **54** is a flexible body that is cantilevered from the core **52**. In particular the flange **54** includes a distal portion **62** (see FIG. 2) that defines the seat surface **58**, and a proximal portion **64** that flexibly couples the distal portion **62** to the core **52** such that the distal portion **62** can move (e.g., pivot) axially upward or downward relative to the core **52**. Moreover, the flange **54** is preferably resiliently biased to the “unloaded configuration” shown in FIGS. 1-3, such that the flange **54** assumes the unloaded configuration when no external load (besides gravity) is acting on and bending the flange **54** relative to the core **52**. It is to be appreciated that the configurations of the flange **54** described below relate to the flange **54** in its unloaded configuration, unless clearly described otherwise.

[0042] As discussed further below, the flexible and resilient configuration of the flange **54** enables the seat surface **58** to flex and move (e.g., pivot) relative to the core **52**, in order to facilitate proper engagement of the seat surface **58** with the nozzle **12** in the closed configuration.

[0043] The flexible and resilient configuration of the flange **54** is enabled by the elastic material used for the disc body **50**, as well as the configuration (e.g., shape, thickness) of the flange **54** as it extends outward from the core **52**.

[0044] More specifically, as shown in FIG. 2, the proximal portion **64** of the flange **54** is tapered such that its axial thickness (measured in the axial direction) gradually increases in the radially outward direction. Meanwhile, the distal portion **62** is substantially larger in axial thickness than the proximal portion **64**. Preferably, a ratio of the smallest axial thickness $t_{sub.1}$ of the proximal portion **64** to the largest axial thickness $t_{sub.2}$ of the distal portion **62** is about 0.20 to about 0.32 and more preferably, about 0.24 to about 0.28. In the present embodiment, the smallest axial thickness $t_{sub.1}$ of the proximal portion **64** is about 0.016 inches and the largest axial thickness $t_{sub.2}$ of the distal portion **62** is about 0.060 inches, such that the ratio of $t_{sub.1}/t_{sub.2}$ is about 0.27.

[0045] The relatively small thickness of the proximal portion **64** enables the flange **54** to bend so the seat surface **58** can move (e.g., pivot) relative to the core **52**. In particular, the distal portion **62** can pivot relative to the core **52** such that the seat surface **58** moves axially upward or downward from the position shown in FIGS. 1-3. Moreover, the relatively larger thickness of the distal portion **62** ensures that the annular shape of the seat surface **58** still has sufficient rigidity for proper alignment with the nozzle **12**. The larger thickness of the distal portion **62** also can increase its weight to help bias the seat surface **58** downward into engagement with the nozzle **12**.

[0046] The core **52**, on the other hand, is significantly larger than the flange **54** in mass and axial thickness. The core **52** is therefore much stiffer than the flange **54**, and can serve as a rigid anchor for the proximal end of the flange **54**. In particular, the core **52** can fit snugly within the disc cavity **28** of the disc holder **26**, and can be coupled to the distal end of the shaft **34**, thereby securing the disc **36** and disc holder **26** thereto. The distal portion **62** of the flange **54** can thus move relative to the core **52** and the other elements **28**, **30** of the disc assembly **22**.

[0047] The disc **36** as described above thus includes a resiliently flexible flange **54** with a seat surface **58** that can pivot and move in the axial direction relative to the core **52**, thereby enabling the seat surface **58** to better conform to the seat surface **44** of the nozzle **12** in the closed configuration, even in low-pressure applications with relatively low biasing forces acting on the disc assembly **22**.

[0048] Preferably, the seat surface **58** will have an axial seat stiffness k relative to the core **52** that is about 500 lb/in to about 1500 lb/in. For the purposes of this disclosure, an “axial seat stiffness” of a seat surface relative to another feature (e.g., a core) corresponds to the ratio of force per unit deformation of the seat surface relative to the other feature in the axial direction. For example, the axial seat stiffness k of the seat surface **58** in the present embodiment can be calculated using the formula below, wherein $d_{sub.0}$ is an axial distance between the seat surface **58** and the bottom of the core **52** in the unloaded configuration when no external load is applied to the seat surface **58**, and $d_{sub.x}$ is an axial distance between the seat surface **58** and the bottom of the core **52** when the distal portion **62** of the disc **36** is forced into engagement with the seat surface **44** in the closed configuration and deflects relative to the core **52**. In the formula below, $P_{sub.x}$ represents the set pressure of the valve **10** (e.g., 50 PSI) that causes the disc assembly **22** to move upward from the closed configuration, while A represents an exposed surface area of the disc **36** that is exposed to fluid within the bore **14** in the closed configuration. In other words, the product of $P_{sub.x}$ and A can be equivalent to and represent the force that is ultimately applied to the seat surface **44** in the closed configuration and causes the seat surface **58** to deflect to the distance $d_{sub.x}$.

$$[00001]k = \frac{P_x * A}{d_0 - d_x}$$

[0049] Notably, the annular groove **60** just above the flange **54** enables the proximal portion **64** to pivot upward without engaging the rest of the disc body **50**. In the present example, the groove **60** is angled such that an outer end of the groove **60** is axially higher than an inner end of the groove **60**. In other examples, the groove **60** may be angled such that its outer end is axially lower than its inner end. Moreover, in some examples, the groove **60** may extend generally perpendicular to the central axis X .

[0050] As the flange **54** engages the seat surface **44** and pivots upward relative to the core **52**, a lower outer edge **66** of the flange **54** can flare radially outward, such that the outer diameter of the flange **54** increases. However, this expansion of the flange's outer diameter could lead to interference between the disc holder **26** and the flange **54**, particularly if the disc cavity **28** has an inner diameter that approximates the outer diameter of the core **52**. Accordingly, in order to prevent such interference, the outer diameter of the flange **54** (in the unloaded configuration) is preferably less than an outer diameter of the core **52**. In particular, a difference between the outer diameter of the flange **54** and the outer diameter of the core **52** can be at least about 0.003 inches.

[0051] Moreover, the flange **54** can include a beveled surface **68** that extends radially outward and upward from an outer perimeter of its annular seat surface **58** to the lower outer edge **66**, wherein the beveled surface **68** extends in a direction oblique to the annular seat surface **58**. In particular, an angle between the beveled surface **68** and a direction of the annular seat surface **58** (which is substantially parallel to the radial direction in the unloaded configuration) is preferably about 10° to about 20°, and more preferably about 15°. This angle of the beveled surface **68** can help further prevent interference between the disc holder **26** and the flange **54** as the lower outer edge **66** flares outward.

[0052] Finally, the disc **36** in the present example defines a lower recess **76** at its bottom end, which can receive fluid from the nozzle **12** and help concentrate fluid force against the disc assembly **22**. The seat surface **58** extends radially outward from an outer perimeter of the lower recess **76**. Moreover, in the present embodiment, a sharp edge **78** is provided between the outer perimeter of the recess **76** and the inner perimeter of the seat surface **58**. However, to help strengthen the edge **78**, the edge **78** can be rounded such that it has a radius of curvature of about 0.020 inches to about 0.040 inches, preferably about 0.030 inches.

[0053] It is to be appreciated that the shape, size, and material of the disc **36** all influence its flexibility and resiliency, and can vary by embodiment without departing from the scope of the disclosure. For instance, FIGS. **4** and **5** show a second embodiment of the relief valve **10** and its disc **36**, wherein the disc **36** is similarly inserted within the disc cavity **28** of the disc holder **26**. Elements in the second embodiment that are similar to those of the first embodiment in FIGS. **1-3** are denoted by the same reference numerals. Except for differences readily apparent from the drawings and description below, it is to be appreciated that the above descriptions of elements for the embodiment in FIGS. **1-3** similarly apply to the elements with the same reference numerals in the second embodiment, and therefore such descriptions have been omitted for the second embodiment.

[0054] In the second embodiment, the disc **36** does not include any groove above its flange **54**. Rather, the core **52** is a substantially cylindrical body and the flange **54** extends radially from a lower end of the core **52**. Moreover, the disc holder **26** in the second embodiment has an annular surface **70** that extends radially between the lower perimeter of the disc cavity **28** and the inner perimeter of the frustoconical skirt surface **30**. The disc **36** is similarly inserted within the disc cavity **28** of the disc holder **26**, but the flange **54** of the disc **36** is arranged outside of the disc cavity **28** and spaced below the annular surface **70**. Accordingly, a gap **72** is provided between the flange **54** and annular surface **70**, which enables the flange **54** to flex upwards without engaging the disc holder **26**.

[0055] The invention has been described with reference to the example embodiments described above. Modifications and alterations will occur to others upon a reading and understanding of this specification. Example embodiments incorporating one or more aspects of the invention are intended to include all such modifications and alterations insofar as they come within the scope of the appended claims.

Claims

1. A disc for a pressure relief valve, the disc comprising a disc body that includes: a core defining a central axis of the disc, and a flange flexibly coupled to the core, the flange defining a seat surface of the disc, wherein the seat surface is axially movable relative to the core.
2. The disc according to claim 1, wherein the flange is resiliently biased toward an unloaded configuration relative to the core, wherein the seat surface extends substantially radial to the central axis in the unloaded configuration.
3. The disc according to claim 1, wherein the disc body comprises polytetrafluoroethylene.
4. The disc according to claim 1, wherein an axial seat stiffness of the flange is about 500 lb/in to about 1500 lb/in.
5. The disc according to claim 1, wherein the flange includes: a distal portion that defines the seat surface, and a proximal portion that flexibly couples the distal portion to the core, wherein a ratio of a smallest axial thickness of the proximal portion to a largest axial thickness of the distal portion is about 0.20 to about 0.32.
6. The disc according to claim 1, wherein the flange extends radially outward from the core.
7. The disc according to claim 1, wherein an outer diameter of the flange is less than an outer diameter of the core, wherein a difference between the outer diameter of the flange and the outer diameter of the core is at least 0.003 inches.
8. The disc according to claim 1, wherein the flange includes a beveled surface that extends radially outward from the seat surface, wherein an angle between a direction of the beveled surface and a direction of the seat surface is about 10° to about 20°.
9. The disc according to claim 8, wherein the flange is resiliently biased toward an unloaded configuration relative to the core, wherein an angle between the beveled surface and a radial direction of the disc in the unloaded configuration is about 10° to about 20°.

- 10.** The disc according to claim 1, wherein the disc body defines an annular groove above the flange, wherein the annular groove is defined at least partially by the flange.
- 11.** The disc according to claim 1, wherein the disc body defines a recess, wherein the seat surface extends outward from an outer perimeter of the recess.
- 12.** The disc according to claim 11, wherein an edge between the outer perimeter of the recess and an inner perimeter of the seat surface has a radius of curvature of about 0.020 inches to about 0.040 inches.
- 13.** A disc assembly comprising: the disc according to claim 1, and a disc holder defining a disc cavity that accommodates the core of the disc.
- 14.** The disc assembly according to claim 13, wherein the disc holder includes a frustoconical skirt surface extending outward from an outer perimeter of the disc cavity.
- 15.** The disc assembly according to claim 13, wherein the disc cavity further accommodates the flange of the disc.
- 16.** The disc assembly according to claim 15, wherein an outer diameter of the flange is less than an outer diameter of the core, wherein a difference between the outer diameter of the flange and the outer diameter of the core is at least 0.003 inches.
- 17.** The disc assembly according to claim 13, wherein the disc holder includes a holder surface that extends outward from an outer perimeter of the disc cavity, wherein an annular gap is defined between the flange of the disc and the holder surface of the disc holder.
- 18.** The disc assembly according to claim 13, the holder surface extending substantially radial to the central axis.
- 19.** The disc assembly according to claim 13, wherein the disc holder includes a frustoconical skirt surface extending outward from an outer perimeter of the holder surface.
- 20.** A pressure relief valve comprising: the disc assembly according to claim 13, and a nozzle defining a bore, an opening at an end of the bore, and a seat surface that surrounds a perimeter to the opening, wherein the disc assembly is translatable between an open configuration in which the seat surface of the disc is spaced from the seat surface of the nozzle, and a closed configuration in which the seat surface of the disc engages the seat surface of the nozzle.
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